

GenMuse: AI-Powered Story Creation Platform

An AI-powered platform that helps anyone turn simple ideas into structured, polished stories.



StoryForge Pro

Mistral-powered story engine

University of
New Haven

University Of New Haven | Data Science | Capstone Project

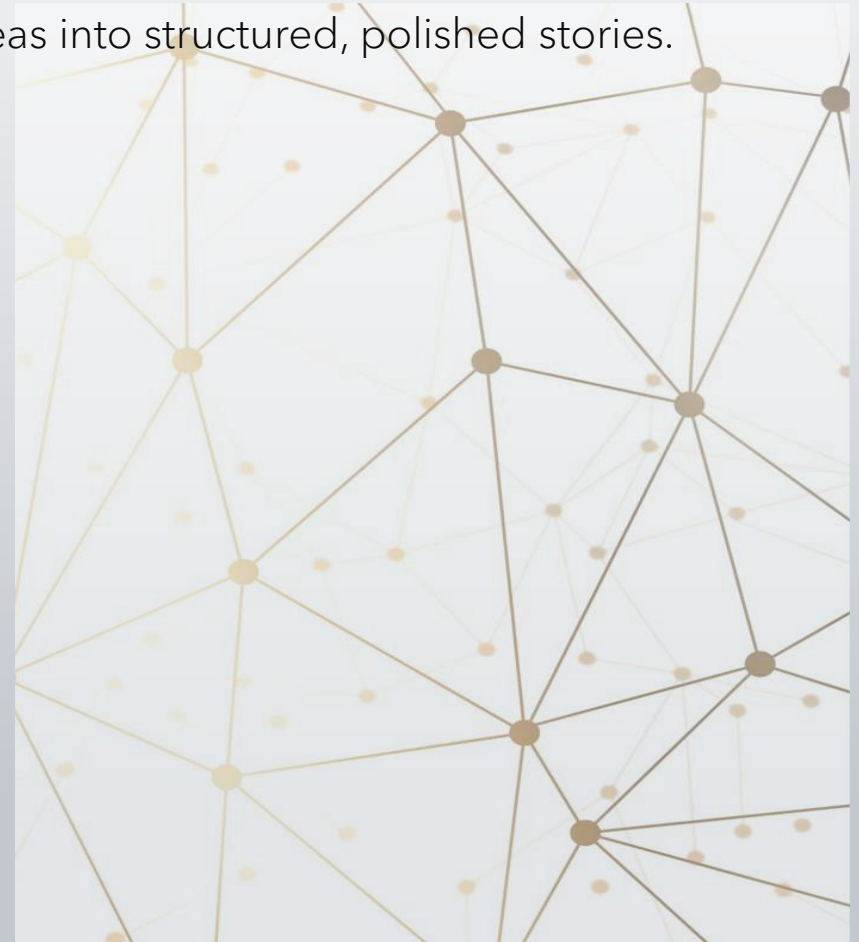
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Problem

Everyone has ideas.

- But turning those ideas into a structured, engaging story is a challenge.
- Writers struggle to expand a simple thought into multiple scenes.
- Editing dialogue or improving pacing takes time and experience.
- Most AI tools just generate long paragraphs with no structure.
- Students and new writers often don't know where to begin.
- What's missing is a tool that acts like a creative partner – not just a text generator.



Our Solution

StoryForge Pro is an AI-powered storytelling companion built around structure, clarity, and creativity.

It helps users:

- Turn an idea into a well-organized multi-scene draft
- Edit each scene individually
- Enhance writing using AI – rewrite, extend, or add dialogue
- Build a world with characters, locations, and timelines
- Convert stories into screenplay format
- Export everything with one click
- A simple idea becomes a full creative workflow.



Key Features

Story Generation

- Your idea becomes a clear, multi-scene storyline – instantly.

Scene-by-Scene Editor

- Edit one scene at a time, the way writers naturally work.

AI Enhancement Tools

- Improve tone and clarity
- Generate natural dialogue
- Continue unfinished scenes

Quick User Accounts

- No complex login – just enough to save your progress.

Powered by Mistral

- Uses the Mistral Chat Completions API to produce high-quality, structured output.



STORYFORGE PRO GENMUSE

USER (BROWSER)

Enters idea, edits
scenes, views story &
screenplay

FRONTEND (REACT + VITE + TAILWIND)

- Story editor UI
- Scene-by-scene editing
- Displays AI outputs
- Generates downloads (JSON/TXT)

REST API Calls

BACKEND (NODE.JS + EXPRESS)

- **BUILDS PROMPTS FOR MISTRAL**
- **HANDLES AUTH, STORY LOGIC, SCENE UPDATES**
- **PROCESSES & CLEANS AI RESPONSE**

OUTPUTS TO FRONTEND:

- Multi-scene Story JSON
- Enhanced Scenes
- Dialogue
- Scene Continuation
- World Model (Characters / Locations / Timeline)
- Screenplay Text

AI Requests /
Responses

MISTRAL API (AI LAYER)

- Generates story
- Rewrites scenes
- Adds dialogue
- Extends narrative
- Extracts world data
- Converts to screenplay

System Architecture

Tech Stack

Frontend

- React
- Vite
- TailwindCSS
- LocalStorage for simple sessions

Backend

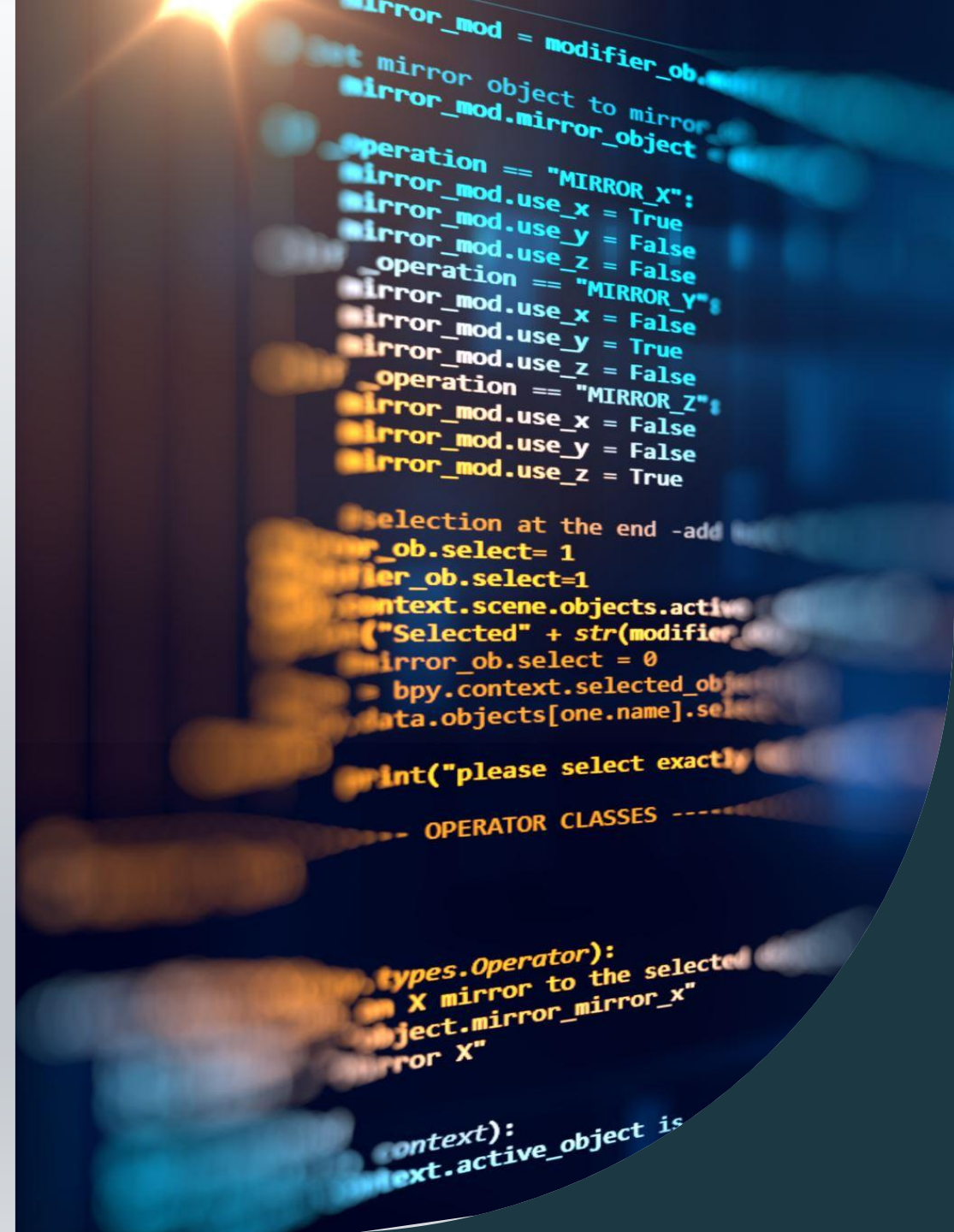
- Node.js
- Express
- REST APIs

AI

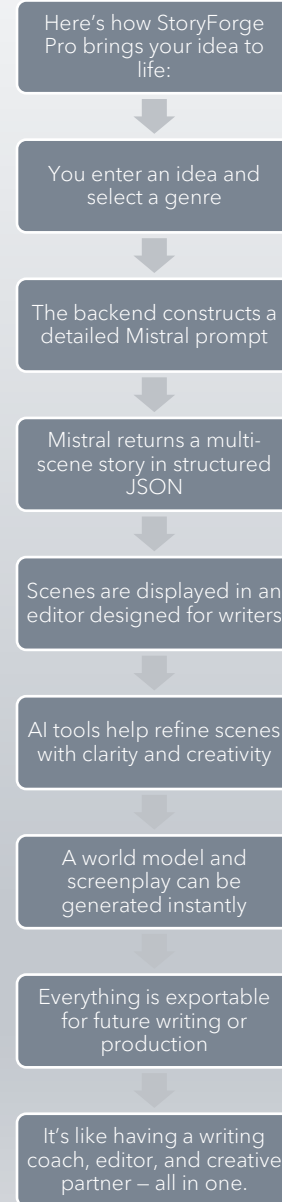
- Mistral Chat Completions API
- Structured JSON prompting

Infrastructure

- Runs locally
- Easy to deploy
- Lightweight and fast



AI Pipeline



Frontend Walkthrough

Login Page

- Simple signup – start creating in seconds.

Dashboard

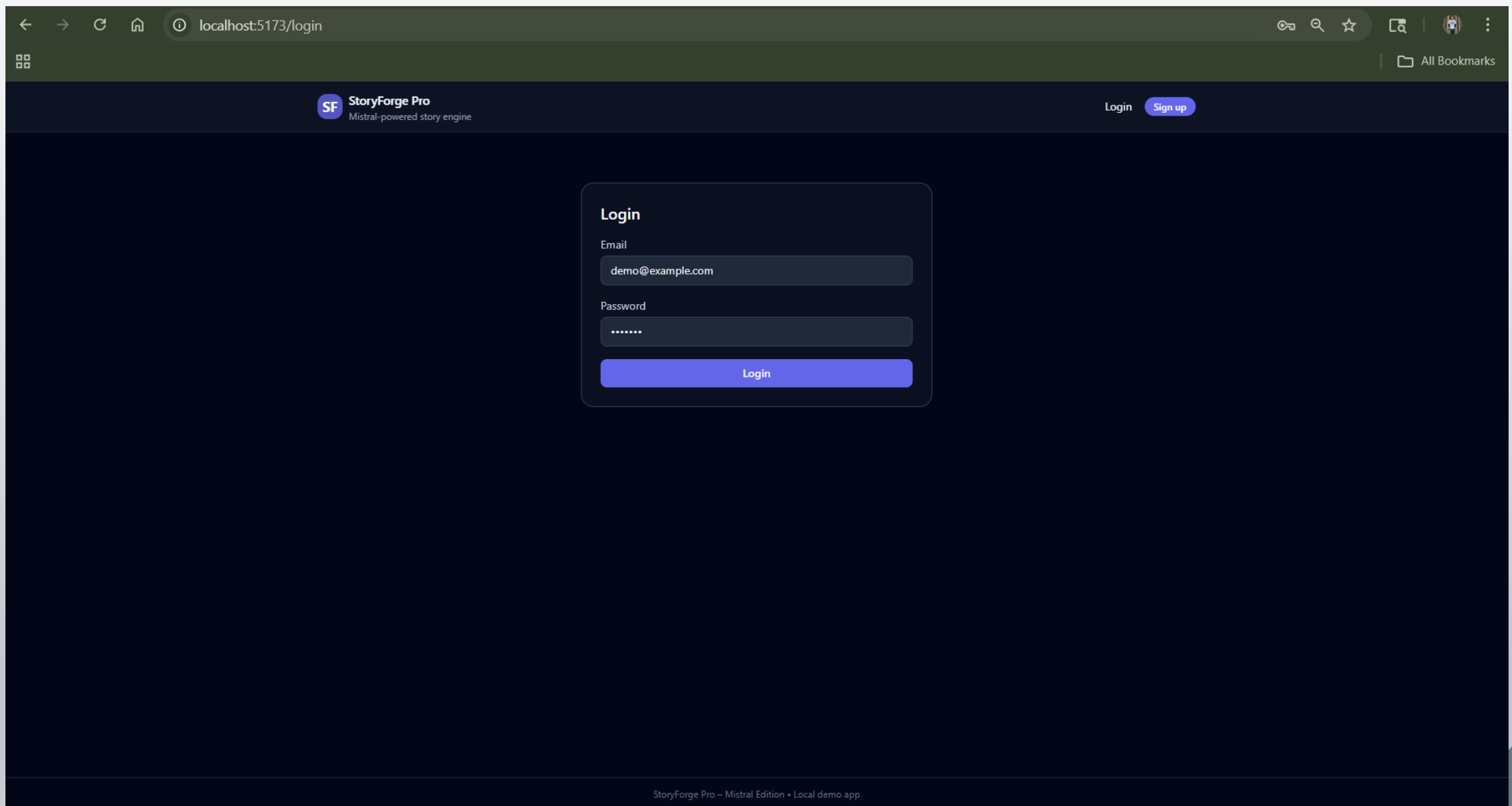
- Enter your idea → choose a genre → click “Generate.”

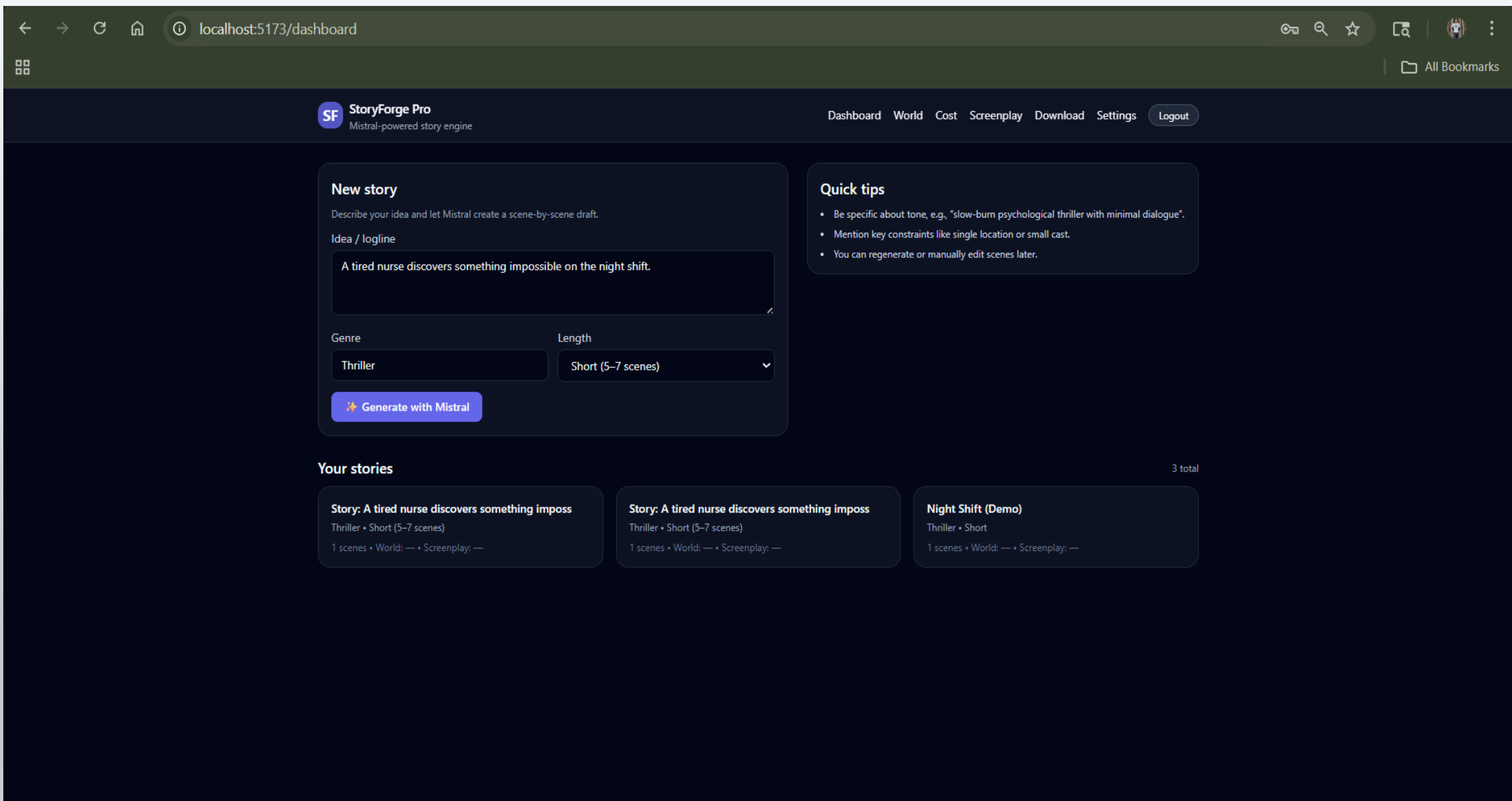
Story Editor

- Improve scenes, generate dialogue, extend moments – all using AI.

Scene Editor

- Focus on one scene at a time. Save versions. Experiment freely.





World builder

Extract characters, locations, and timeline from your story using Mistral.

🔍

Extract from story

Characters

Captain Elara Vey

Rogue pilot

A rogue pilot with a ship full of secrets, known for her exceptional flying skills and resourcefulness.

Governor Kael

Leader of New Eden

The leader of the hidden human colony New Eden, who has kept the colony safe from the empire for centuries.

Imperial Commander Vex

Imperial adversary

A ruthless and cunning commander of the imperial fleet, determined to capture Elara and uncover the secret of New Eden.

Rebel Leader Jax

Freedom fighter

A former imperial officer turned rebel leader, who joins forces with Elara to ignite a rebellion against the empire.

Locations

Deep Space

The vast, empty expanse of space where Elara drifts with her ship, the Stellar Wanderer.

New Eden

A hidden human colony located in a remote star system, thriving behind a dense asteroid field.

Asteroid Field

A dense field of asteroids that hides New Eden from the empire and serves as a battleground for Elara and her pursuers.

Timeline

Day 1

- Captain Elara Vey drifts through deep space and picks up a faint, repeating human signal.

Day 2

- Elara traces the signal to a remote star system and discovers the hidden colony of New Eden.

Day 3

- Elara is welcomed by Governor Kael and learns about the colony's history and the threat of the empire.

Day 4

- Elara flees New Eden, leading the imperial fleet away from the colony.

Day 5

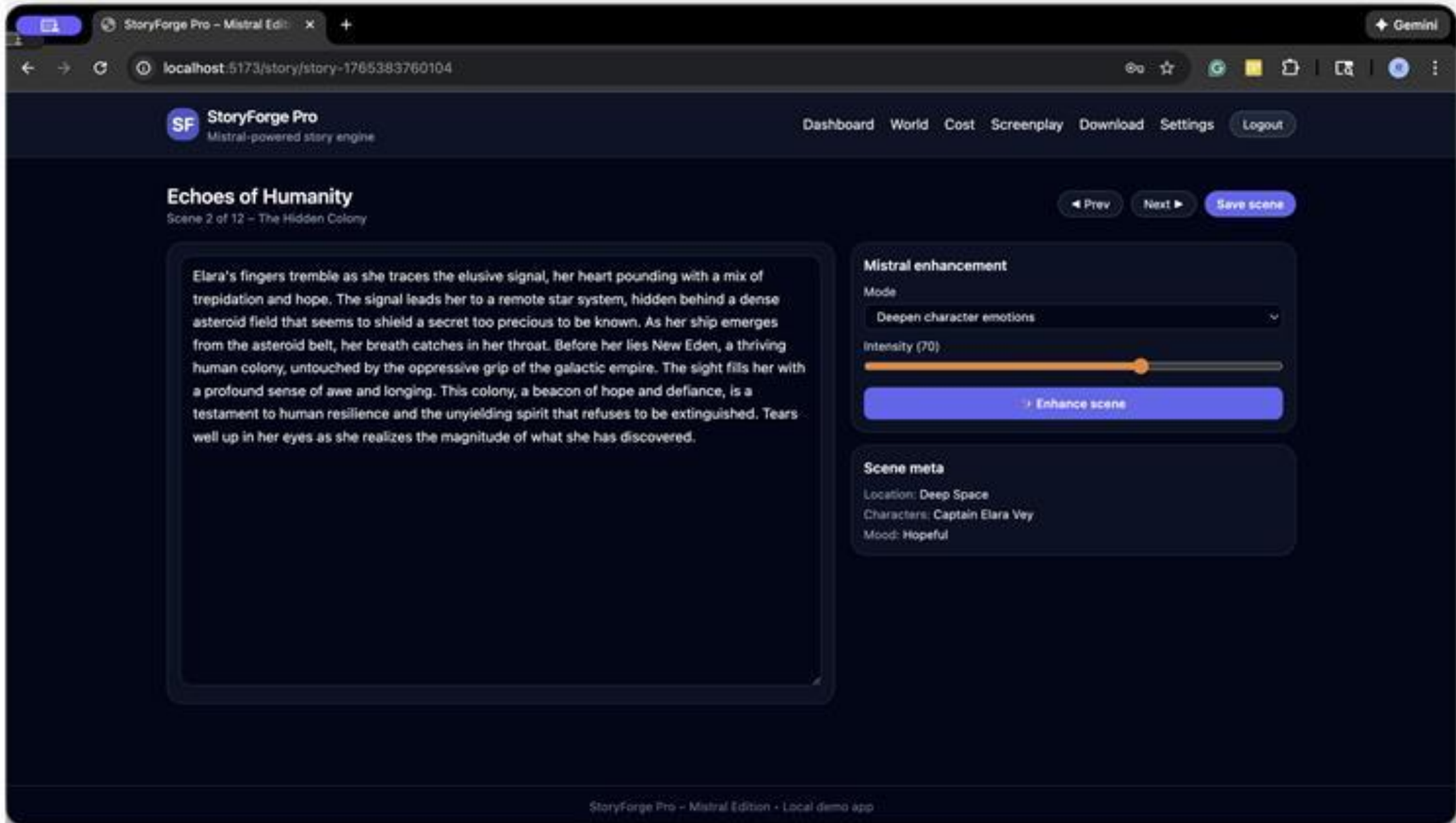
- Elara faces a crucial decision: continue evading the empire or ignite a rebellion.

Day 6

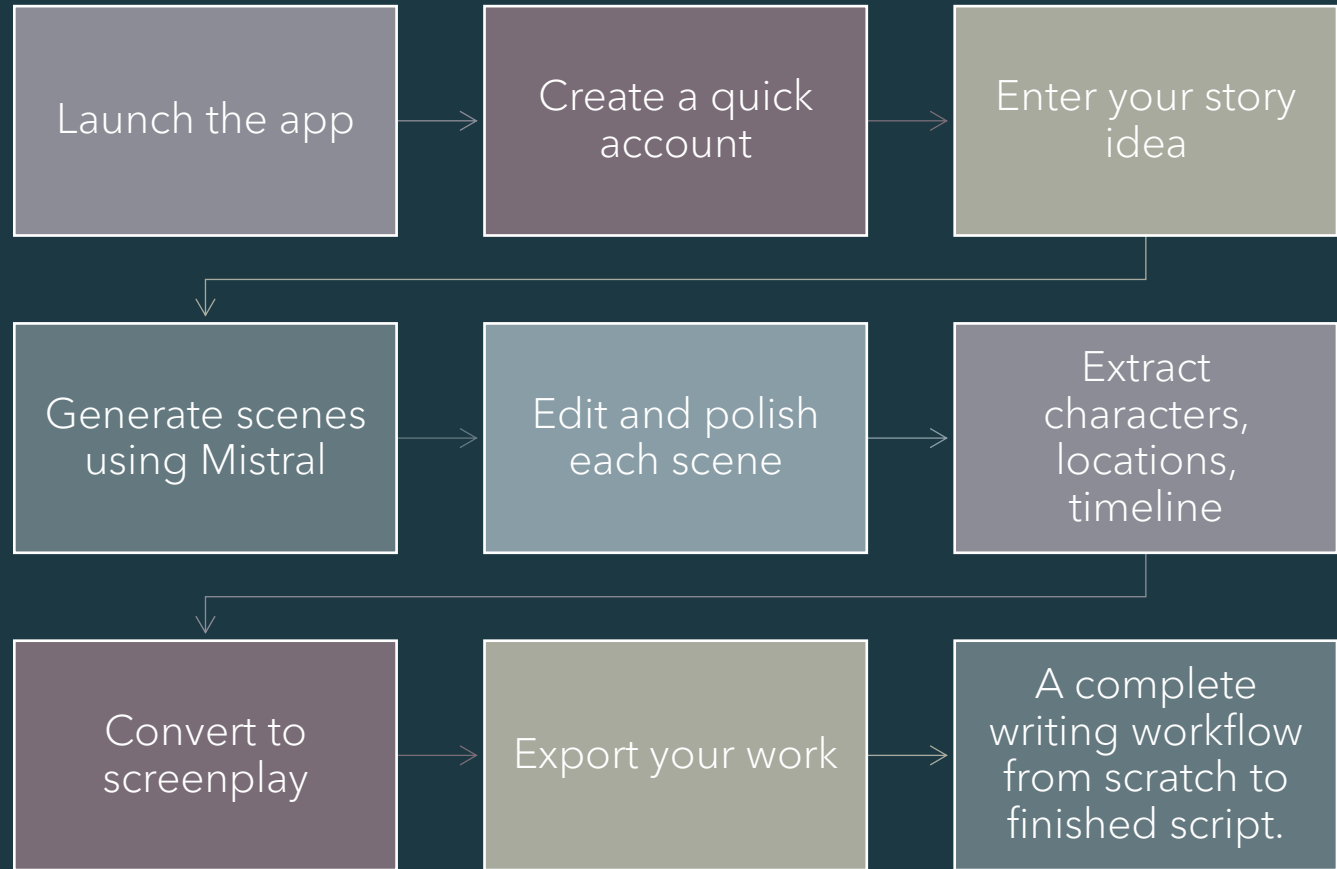
- Elara chooses to fight and contacts Rebel Leader Jax to devise a plan to expose the empire's atrocities.

Day 7

- The empire launches a full-scale assault on New Eden, and Elara and her allies fight to defend the colony.



Demo Flow



Strengths

- Structured output, no more messy AI text
- Human-centric editing workflow
- Fast, modern, cinematic UI
- Full creative pipeline in one tool
- Built for both new writers and experienced storytellers
- Mistral provides consistent, high-quality writing



Future Enhancements

StoryForge Pro has room to grow into a full storytelling studio:

- AI image generation
- AI voice narration
- Collaborative multi-user writing
- Full version history and revision tracking
- Automated storyboarding
- Mobile app version
- Advanced publishing integrations



PROJECT VIDEO

AI Studio - Mistral AI

StoryForge Pro - Mistral Editi

+

localhost:5173/dashboard

SF StoryForge Pro

Mistral-powered story engine

DashboardWorldCostScreenplayDownloadSettingsLogout

New story

Describe your idea and let Mistral create a scene-by-scene draft.

Idea / logline

Beneath the melting polar ice, a research team uncovers a submerged library filled with books that write themselves. But the stories begin predicting their deaths—one by one.

Genre

Thriller

Length

Medium (8–12 scenes)

Generate with Mistral

Quick tips

Be specific about tone, e.g., “slow-burn psychological thriller with minimal dialogue”.

Mention key constraints like single location or small cast.

You can regenerate or manually edit scenes later.

Your stories

6 total

Story: Title: The Drowned Archive Prompt: Bene

Science Fiction • Medium (8–12 scenes)

1 scenes • World: — • Screenplay: —

Story: Title: The Drowned Archive Prompt: Bene

Science Fiction • Medium (8–12 scenes)

1 scenes • World: — • Screenplay: —

From Hyderabad to the Podium

Drama • Medium

12 scenes • World: ✓ • Screenplay: ✓

Story: A story of a boy from Hyderabad who is f

Thriller • Feature (13–20 scenes)

1 scenes • World: — • Screenplay: —

The Nameless Heir

Drama • Short

7 scenes • World: ✓ • Screenplay: ✓

Night Shift (Demo)

Thriller • Short

1 scenes • World: — • Screenplay: —

Why This Matters?

Good stories shape communication, entertainment, and education. But not everyone knows how to turn an idea into something meaningful.

StoryForge Pro:

- Lowers the barrier to writing
- Helps students and creators learn structure
- Makes storytelling enjoyable, not intimidating
- Empowers anyone to bring imagination to life

Summary

GenMuse is:

- A creative partner
- A structured writing assistant
- An AI-powered editor
- A world-building tool
- A screenplay generator
- It helps anyone write with less effort and more imagination.

A hand holding a lit sparkler against a background of colorful bokeh lights. The sparkler is bright and glowing, with many small sparks flying out. The background is dark with various colored circles (bokeh) in shades of orange, red, blue, and green. The overall mood is celebratory and warm.

Thank You

Every great story begins with a spark....